1-3 Trigonometric Functions

師大工教一

Degrees vs. radians

1. $s = r\theta$ (θ in radians)

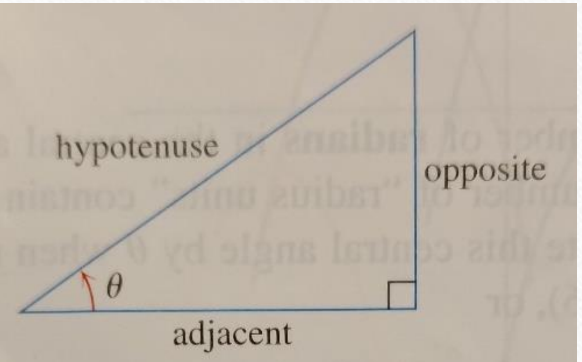
2. $\pi = 180^{\circ}$

度度量

弳度量

↓

$S = r\theta$



$$\sin\theta = \frac{\text{opp}}{\text{hyp}} \qquad \csc\theta = \frac{\text{hyp}}{\text{opp}}$$

$$\cos\theta = \frac{\text{adj}}{\text{hyp}} \qquad \sec\theta = \frac{\text{hyp}}{\text{adj}}$$

$$\tan\theta = \frac{\text{opp}}{\text{adj}} \qquad \cot\theta = \frac{\text{adj}}{\text{opp}}$$

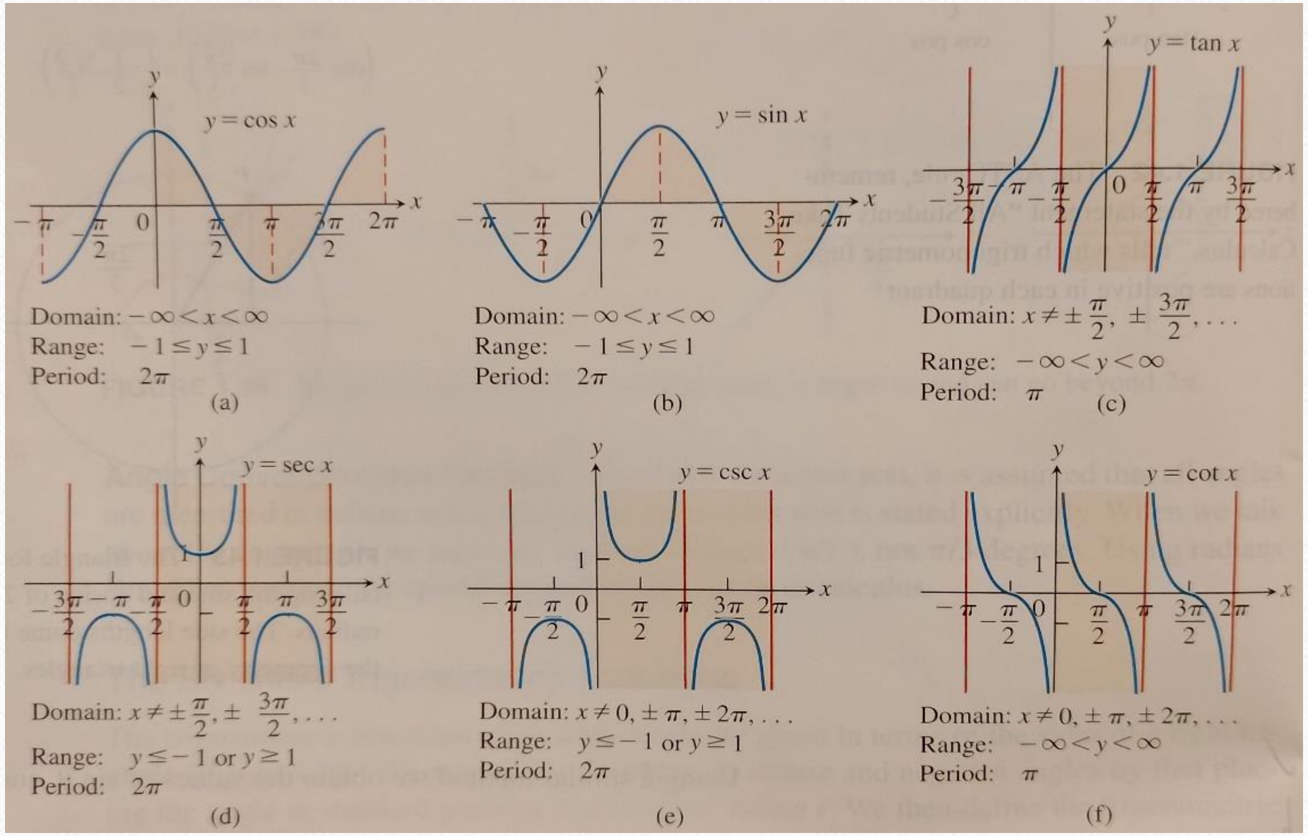
0	**30**	**45**	**60**	**90**
0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
0	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$	

Definition A function $f(x)$ is periodic if there is a positive number p such that $f(x+\boxed{p})=f(x)$ for every value of x. The smallest of p is the period of f.

週期

periodic functions 週期函數

period 週期



y = cos x
Domain: −∞ < x < ∞
Range: −1 ≤ y ≤ 1
Period: 2π
(a)
y = sin x
Domain: −∞ < x < ∞
Range: −1 ≤ y ≤ 1
Period: 2π
(b)
y = tan x
Domain: x ≠ ± π/2, ± 3π/2, . . .
Range: −∞ < y < ∞
Period: π
(c)
y = sec x
Domain: x ≠ ± π/2, ± 3π/2, . . .
Range: y ≤ −1 or y ≥ 1
Period: 2π
(d)
y = csc x
Domain: x ≠ 0, ± π, ± 2π, . . .
Range: y ≤ −1 or y ≥ 1
Period: 2π
(e)
y = cot x
Domain: x ≠ 0, ± π, ± 2π, . . .
Range: −∞ < y < ∞
Period: π
(f)

Trigonometric Identities

1. $\sin^2\theta + \cos^2\theta = 1, 1 + \tan^2\theta = \sec^2\theta, 1 + \cot^2\theta = \csc^2\theta$

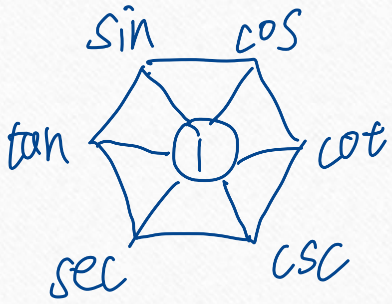
2. Addition Formulas
$$\cos(A+B) = \cos A\cos B - \sin A\sin B$$
$$\sin(A+B) = \sin A\cos B + \cos A\sin B$$

3. Double-Angle Formulas
$$\cos 2\theta = \cos^2\theta - \sin^2\theta$$
$$\sin 2\theta = 2\sin\theta\cos\theta$$

4. Half-Angle Formulas
$$\cos^2\theta = \frac{1+\cos 2\theta}{2}$$
$$\sin^2\theta = \frac{1-\cos 2\theta}{2}$$

5. Law of Cosines $c^2 = a^2 + b^2 - 2ab\cos C$

6. Special Inequality
$$-|\theta| \le \sin\theta \le |\theta|$$
$$-|\theta| \le 1 - \cos\theta \le |\theta|$$



HW1-3

- **No HW**

1-5 Exponential Functions

師大工教一

Definition: Function $f(x)=a^x, a>0$, is the exponential function with base a.

指數 以a為底

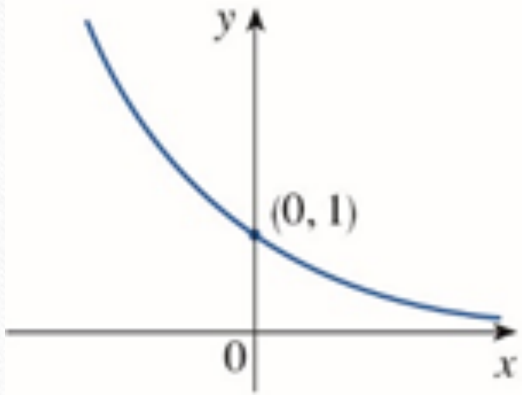
Recall: $n\in\mathbb{N}, a^n = a\cdot a\cdot\cdots\cdot a$

$$a^0=1$$

$$a^{-n}=\frac{1}{a^n}$$

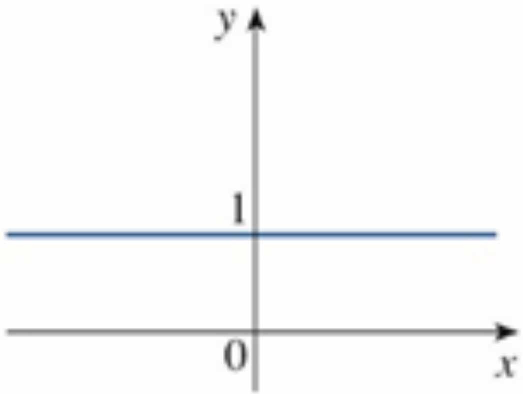
$$a^{\frac{1}{n}}=\sqrt[n]{a}, a^{\frac{p}{q}}=\sqrt[q]{a^p}$$

$$\sqrt{3}=1.7320508\cdots, 2^1, 2^{1.7}, 2^{1.73}, 2^{1.732}, 2^{1.7320}, 2^{1.73205}, \cdots \to 2^{\sqrt{3}}$$



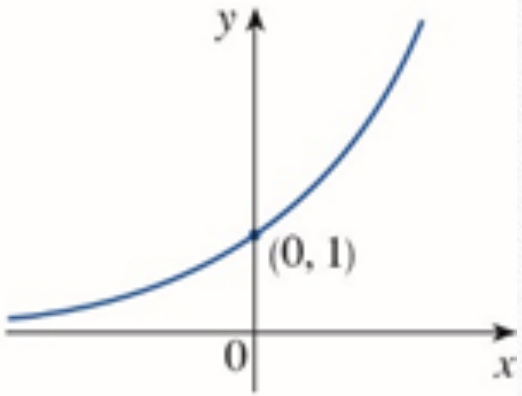
$0 < a < 1$

$y = a^x$



$a = 1$

$y = a^x$



$a > 1$

$y = a^x$

※Laws of Exponents(p35)

指數律

$a,b>0$ and $x,y\in\mathbb{R}$

1. $b^x\cdot b^y=b^{x+y}$

2. $\dfrac{b^x}{b^y}=b^{x-y}$

3. $\left(b^x\right)^y=b^{xy}$

4. $a^x\cdot b^x=\left(ab\right)^x$

※The Natural Exponential Function(p54)

e：尤拉數

$y=e^x=\exp\left(x\right)$, where e is an irrational number, $e=2.718281828\cdots$

HW1-5

- **No HW**

1-6 Inverse Functions
反 函數
and Logarithms
對數

師大工教一

※Inverse Functions

反函數

Definition: A function $f(x)$ is **one-to-one** if $f(x_1) \neq f(x_2)$ whenever $x_1 \neq x_2$.

Definition: Suppose that $f(x)$ is a one-to-one function with domain A and range B. The **inverse function** f^{-1} has domain B and range A and is defined by $f^{-1}(y) = x \Leftrightarrow f(x) = y$.

合成 function | EX:

$f(x) = x+2$, $g(x) = x^2$

$(f \circ g)(x) = f(g(x)) = f(x^2)$

circle

$= x^2 + 2$

$(g \circ g)(x) = g(g(x)) = g(x^2)$

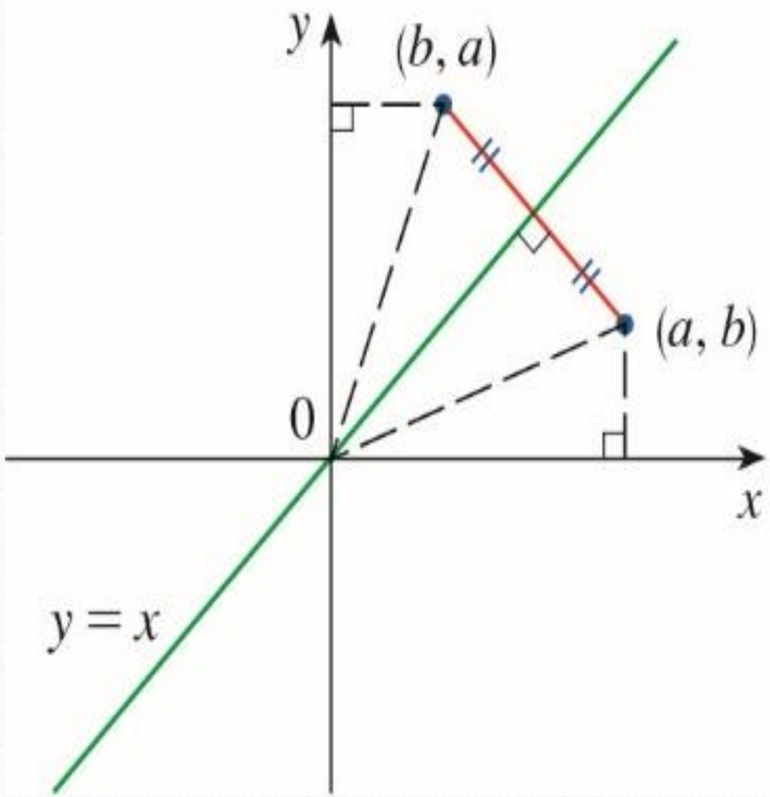
$= (x^2)^2 = x^4$

Note: $\boxed{f^{-1}(x)}$ does not mean $\dfrac{1}{f(x)}$.

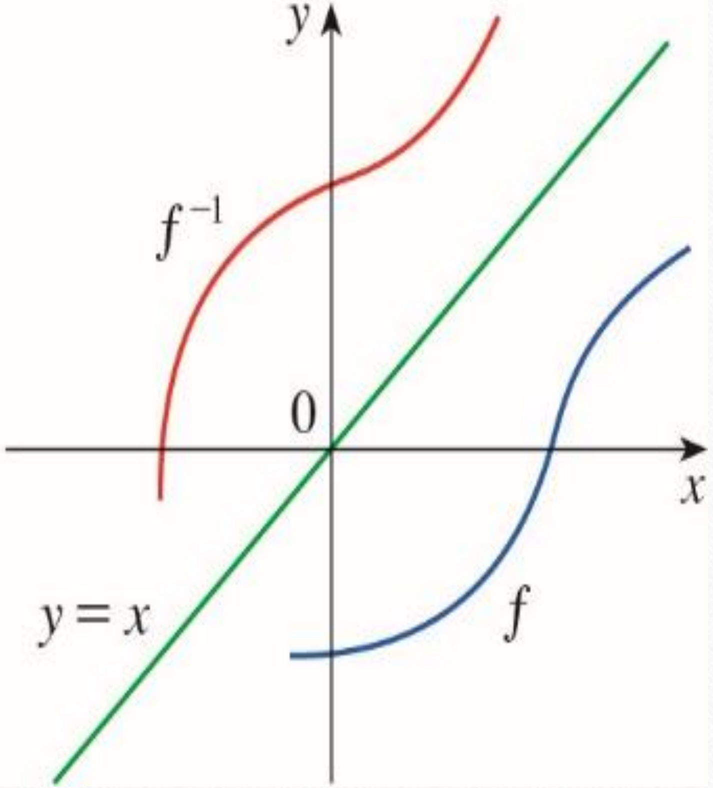
讀作：f inverse

Cancellation Equations: $f^{-1}(f(x)) = x, x \in A$, $f(f^{-1}(y)) = y, y \in B$

消去律



y
(b, a)
(a, b)
0
x
y = x

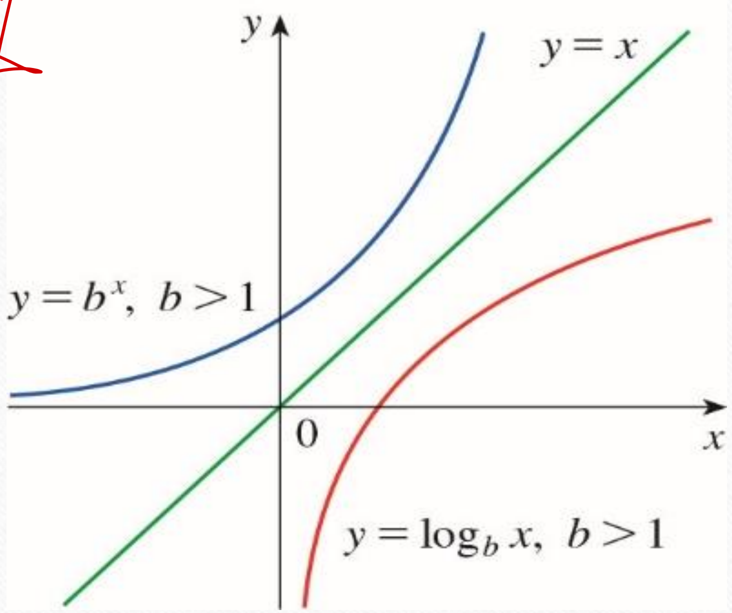


y
f^{-1}
0
x
y = x
f

※Logarithmic Functions

Definition: The logarithmic function with base b, written $y = \log_b x$, is the inverse of the base b exponential function $y = b^x, (b > 0, b \neq 1)$.

$y = \log_b x \Leftrightarrow y = b^x$



The natural logarithmic function $y = \log_e x = \ln x$

The common logarithmic function $y = \log_{10} x = \log x$

※Properties of Logarithms

1. $\ln(bx) = \ln b + \ln x$ 2. $\ln\left(\frac{b}{x}\right) = \ln b - \ln x$ 3. $\ln x^r = r \ln x$

※4. $a^{\log_a x} = x, \log_a a^x = x, e^{\ln x} = x, \ln e^x = x$

※5. $a^x = \left(e^{\ln a}\right)^x = e^{\ln a \cdot x}$

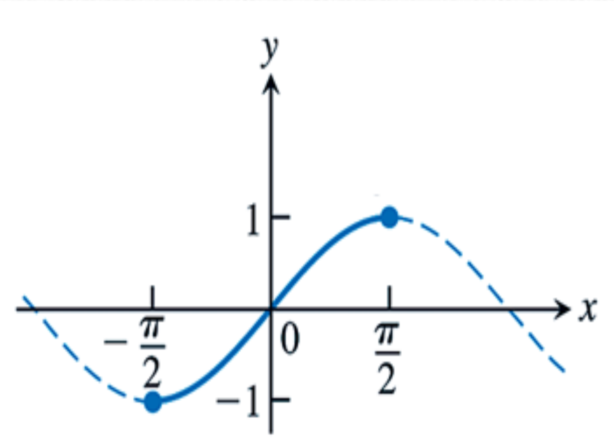
6. $\log_a x = \dfrac{\ln x}{\ln a}$ $\log_a x = \dfrac{\log_b x}{\log_b a}$

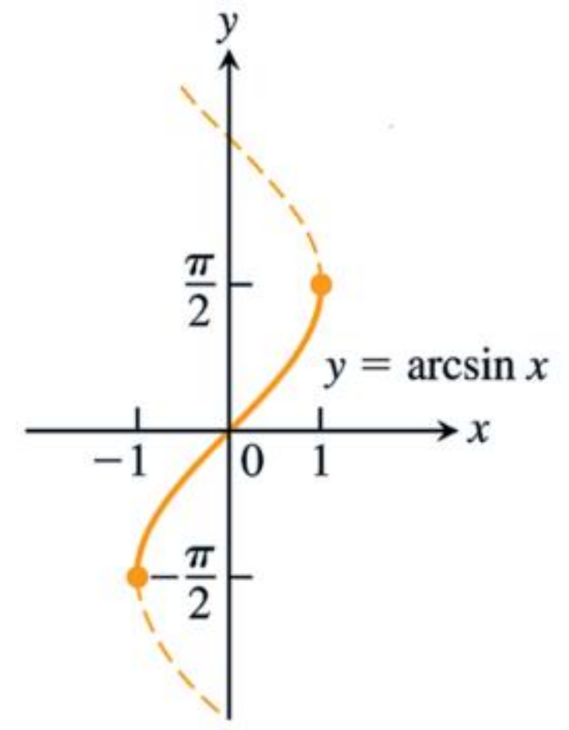
※Inverse Trigonometric Functions

$$\sin\frac{\pi}{6}=\frac{1}{2}\Leftrightarrow \sin^{-1}\left(\frac{1}{2}\right)=\frac{\pi}{6}=\arcsin\left(\frac{1}{2}\right)=\frac{\pi}{6}$$

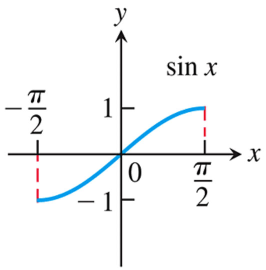
Definition: $\sin\theta = a \Leftrightarrow \sin^{-1} a = \theta$, etc.

Properties: 1. $\sin\left(\sin^{-1} a\right)=a$, 2. $\sin^{-1}\left(\sin\theta\right)=\theta$





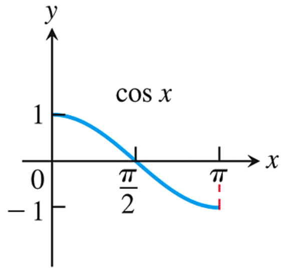
Domain restrictions that make the trigonometric functions one-to-one



$y = \sin x$

Domain: $[-\pi/2, \pi/2]$

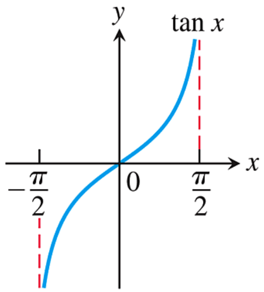
Range: $[-1, 1]$



$y = \cos x$

Domain: $[0, \pi]$

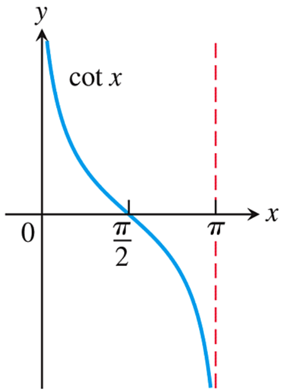
Range: $[-1, 1]$



$y = \tan x$

Domain: $(-\pi/2, \pi/2)$

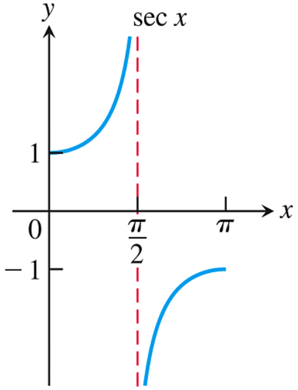
Range: $(-\infty, \infty)$



$y = \cot x$

Domain: $(0, \pi)$

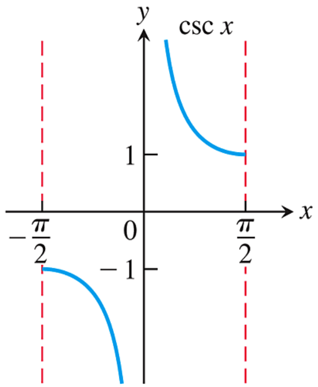
Range: $(-\infty, \infty)$



$y = \sec x$

Domain: $[0, \pi/2) \cup (\pi/2, \pi]$

Range: $(-\infty, -1] \cup [1, \infty)$



$y = \csc x$

Domain: $[-\pi/2, 0) \cup (0, \pi/2]$

Range: $(-\infty, -1] \cup [1, \infty)$

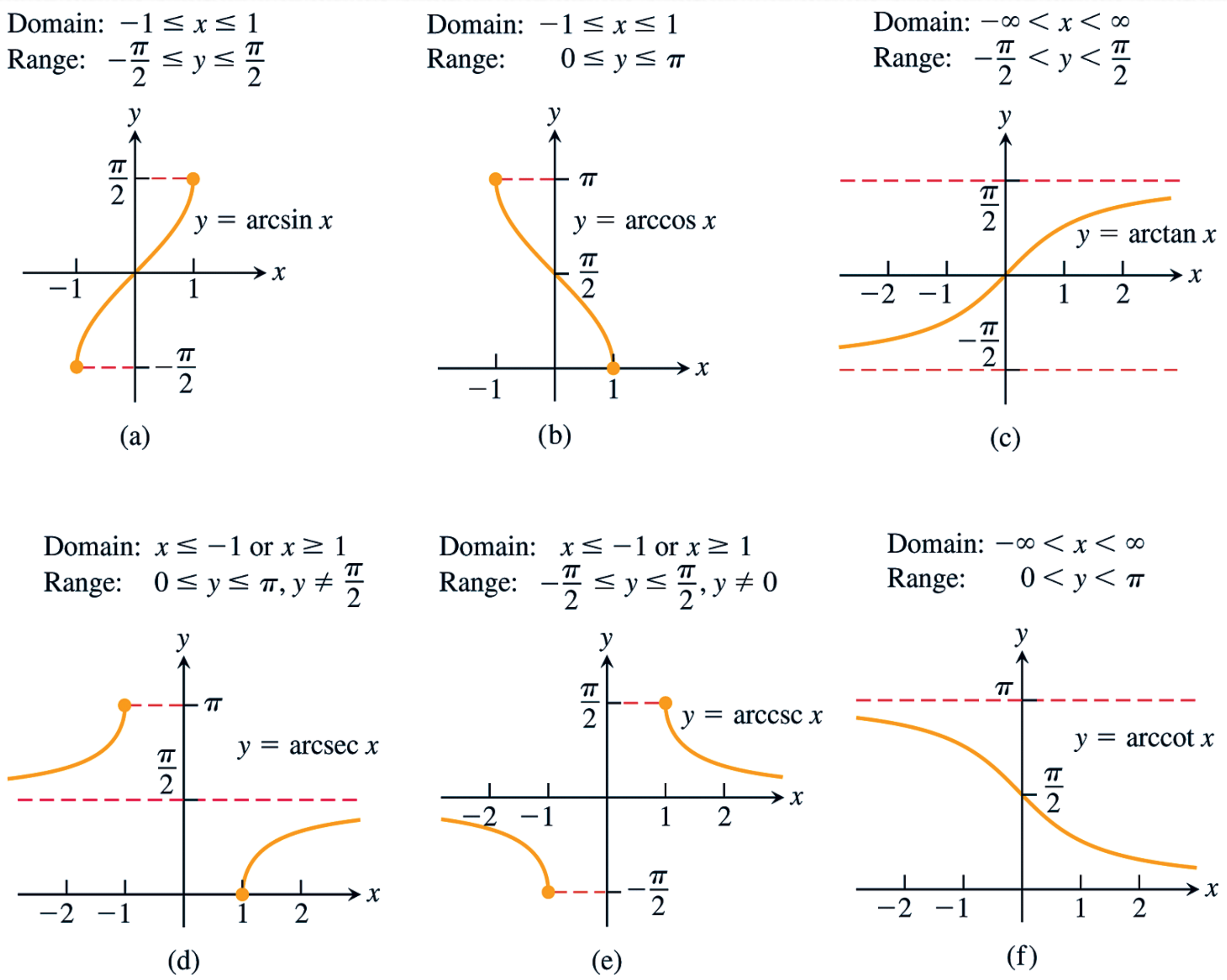


FIGURE 1.64 Graphs of the six basic inverse trigonometric functions.

HW1-6

- **HW: 14,60,62,71**

111 1. (3 pts) Evaluate $\sin(2\sin^{-1}(-\frac{3}{5}))$. (note : $\sin^{-1} x = \arcsin x$)

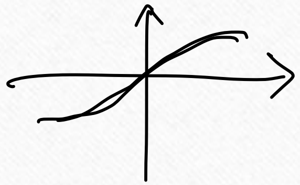
求

Let $\sin^{-1}(-\frac{3}{5}) = \theta$

$\Rightarrow \sin(\sin^{-1}(-\frac{3}{5})) = \sin\theta$

$\sin\theta = -\frac{3}{5}$

$\cos\theta = \frac{4}{5}$



$-\frac{\pi}{2} \le \theta \le \frac{\pi}{2}$

$\sin(2\theta)$

$= 2\sin\theta\cos\theta$

$= 2 \times -\frac{3}{5} \times \frac{4}{5}$

$= \frac{-24}{25}$ #